The Golden Signature Engine

TRACK: Option B – Optimization Engine Focus

AI-driven, multi-objective optimization platform designed directly for manufacturing process engineers. It analyzes historical batch production and process data to find the optimal machine configuration (the "Golden Signature") that minimizes energy consumption while maximizing yield and strictly adhering to quality constraints.

The Working Prototype

Functionality-

The application acts as a predictive and prescriptive analytics layer over existing SCADA or MES (Manufacturing Execution Systems). It ingests real-time and historical sensor data and allows plant operators to dynamically shift priorities (e.g., dialing up "Energy Focus" if power costs spike or "Yield Focus" if demand is high). The engine mathematically guarantees that the suggested parameters will not severely degrade interconnected variables like Content Uniformity or Dissolution Rate.

Key Features

- **Adaptive Priority Sliders–**
  Operators control the optimization weights directly via the UI, removing the need to ask data scientists for entirely new models when business goals change.

- **Explainable AI (XAI) & Feature Importance–**
  A visual breakdown generated from the core Machine Learning model that shows exactly which machine parameters (e.g., Machine Speed vs. Drying Temp) govern the final physical outcomes of the batch.

- **Digital Twin "What–If" Simulator–**
  If an operator cannot achieve the exact Golden Signature recommended by the AI, they can manually override the sliders in the interface. The embedded Digital Twin instantly predicts the new energy consumption, quality, and yield impacts in real–time.

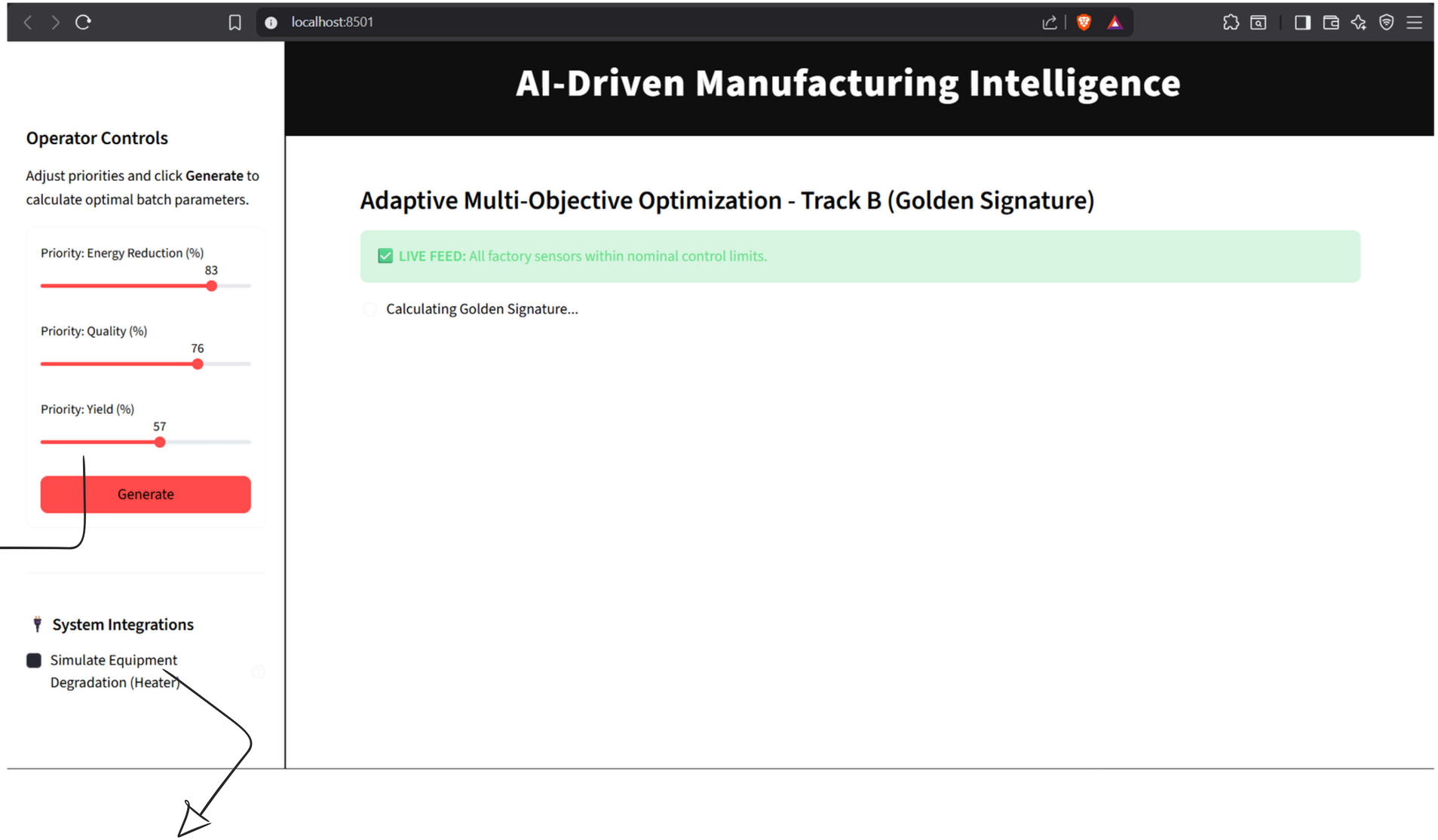
Key Features

- **Prescriptive Maintenance Anomaly Engine–**
  Unlike standard predictive maintenance that simply raises an alert (e.g., "Heater failing"), the application is prescriptive. If an anomaly is toggled, it automatically recalculates a new, cooler Golden Signature that compensates for the degraded equipment, preventing batch failure.

- **21 CFR Part 11 Compliant Audit Logging–**
  Secure, immutable logs track every time an operator deploys an AI-recommended signature to the EDGE controller.

- **Cost & Carbon Calculator–**
  Immediately translates kWh energy savings into direct dollar savings and pounds of carbon emissions avoided.

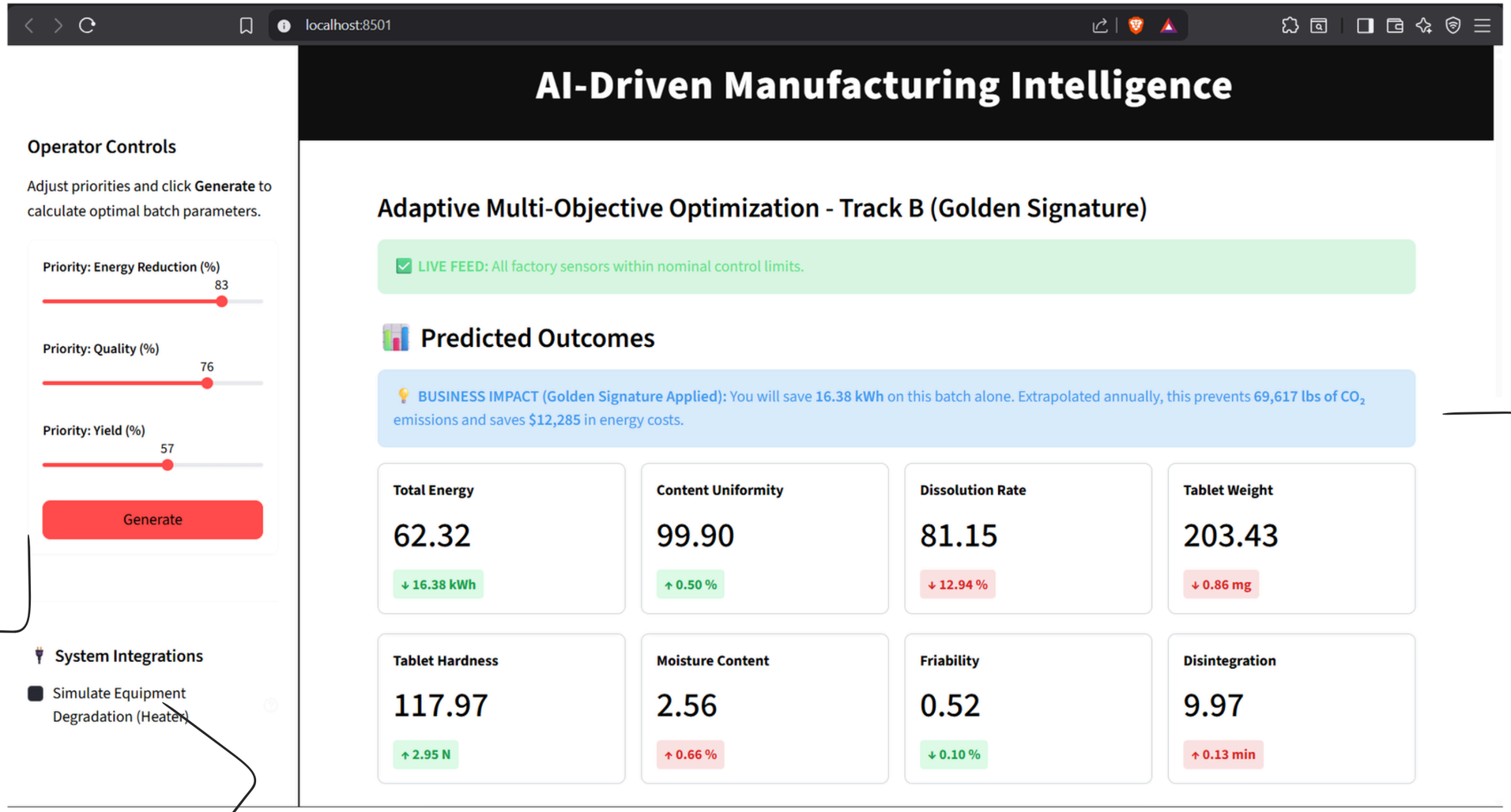
Prototype Walkthrough



Adaptive Priority Sliders

Prescriptive Maintenance Anomaly Engine

Prototype Walkthrough

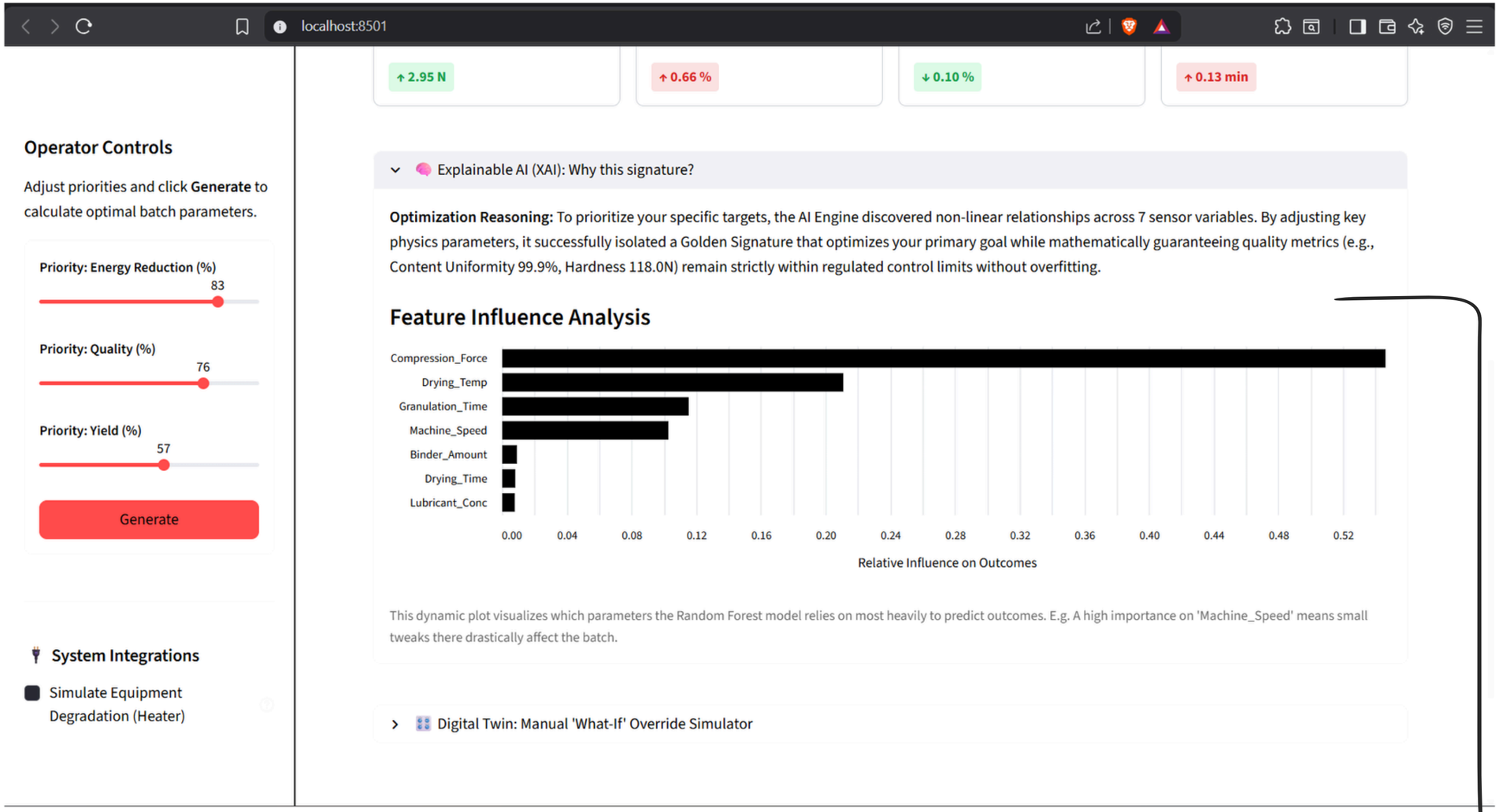


Adaptive Priority Sliders

Prescriptive Maintenance Anomaly Engine

Cost & Carbon Calculator

Prototype Walkthrough



Explainable AI (XAI) & Feature Importance

Prototype Walkthrough

localhost:8501

AI-Driven Manufacturing Intelligence

Comparison Table

System has identified a more efficient parameter set (Golden Signature) based on your current priorities.

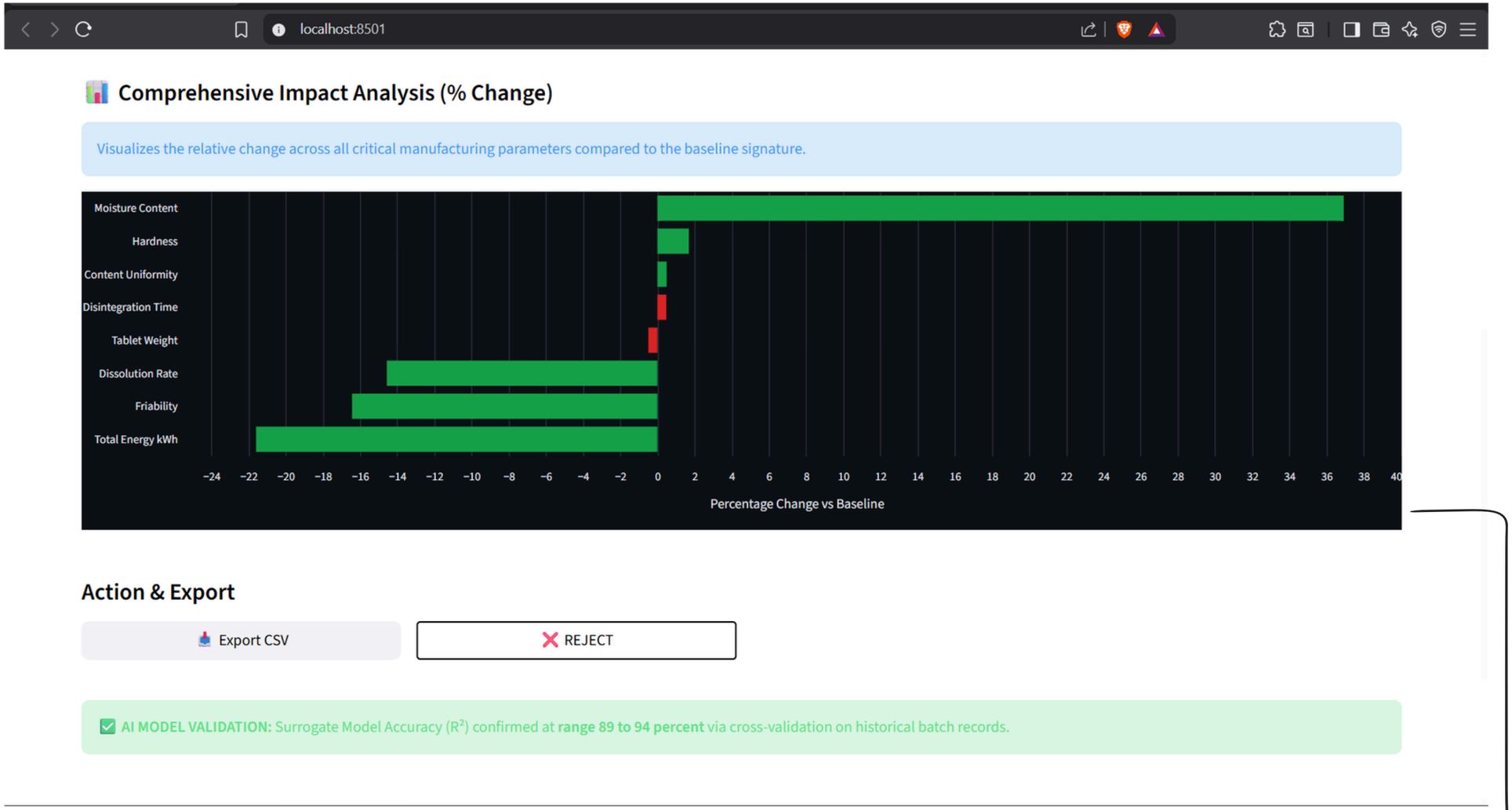
Parameter	Previous Value	Current (Golden) Value	% Change
Granulation_Time	25	28.74	+14.96%
Binder_Amount	10.5	11.61	+10.57%
Drying_Temp	80	46.88	-41.40%
Drying_Time	120	66.5	-44.58%
Compression_Force	18	15.49	-13.96%
Machine_Speed	60	20.42	-65.97%
Lubricant_Conc	2	0.65	-67.43%

Comprehensive Impact Analysis (% Change)

Visualizes the relative change across all critical manufacturing parameters compared to the baseline signature.

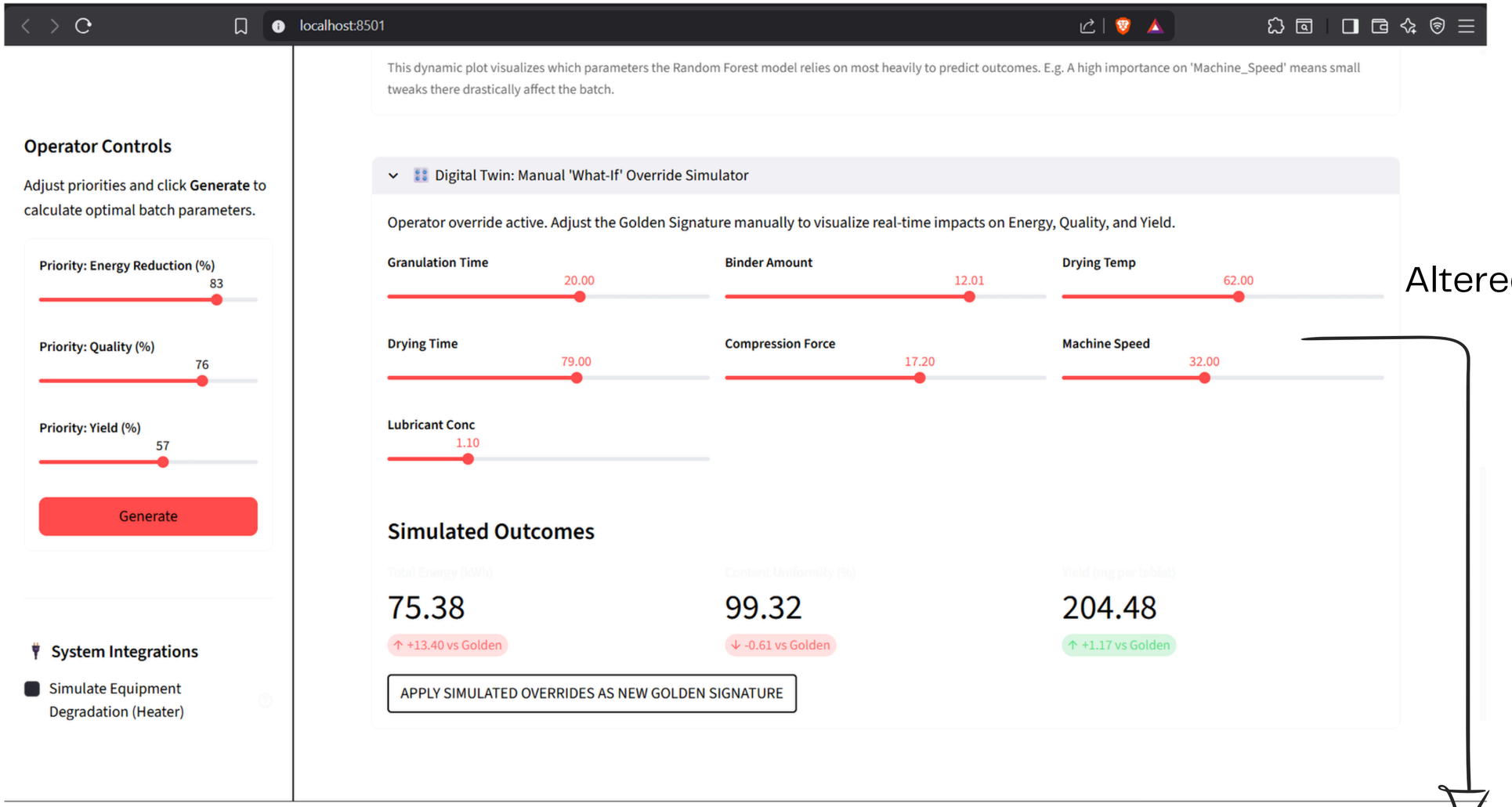
Compares default values with Golden Signature Values

Prototype Walkthrough



Comprehensive Impact Analysis

Prototype Walkthrough

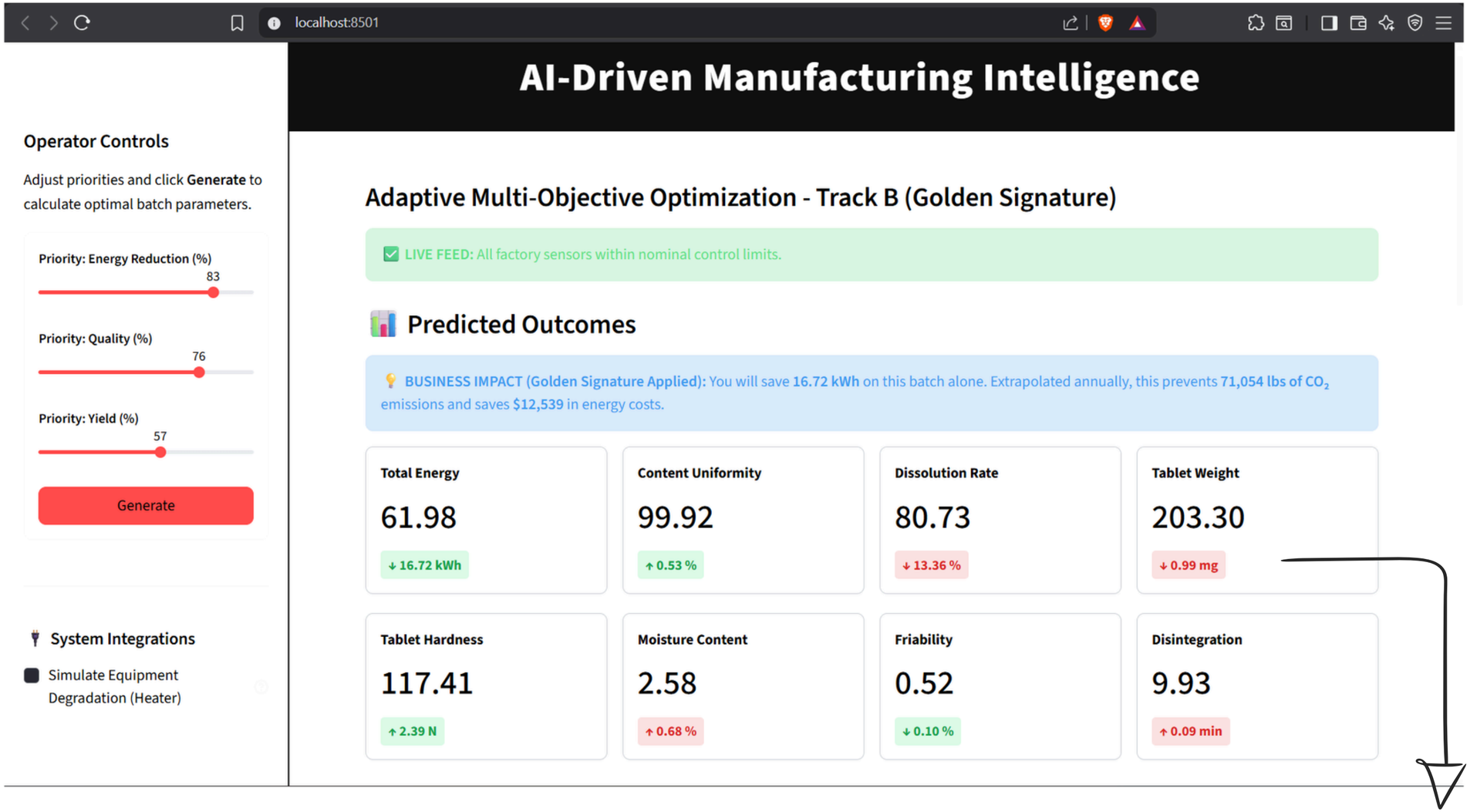


Altered values

Digital Twin "What-If" Simulator

If an operator cannot achieve the exact Golden Signature recommended by the AI, they can manually override the sliders in the interface. The embedded Digital Twin instantly predicts the new energy consumption, quality, and yield impacts in real-time.

Prototype Walkthrough



Predicted Outcomes will be updated if Operator alter values from **Digital Twin "What–If" Simulator**

Technical Architecture

The system is built on a modular, decoupled architecture

- **Data Integration Pipeline (data_pipeline.py)-**
  Ingests and merges granular time-series process data with summary production data. It aggregates 1-minute time steps, calculates total batch energy (integral of power over time),and prepares the unified dataset.

- **Machine Learning Surrogate Model (surrogate_model.py)-**
  Acts as the "Digital Twin". It uses a multi-target RandomForestRegressor. Since the Hackathon dataset only provided one matching batch, the engine utilizes a custom physics-informed data augmentation loop to generate synthetic, noisy relationships to train the model realistically without overfitting.

Technical Architecture

The system is built on a modular, decoupled architecture

- **Global Optimization Engine (optimization_engine.py)–**
  The core intelligence. It uses the Surrogate Model as a simulated factory. It runs a global search algorithm against a customized penalty reward function to find the absolute minimum energy state without sacrificing quality boundaries.

- **Operator Interface (app.py)–**
  A high contrast, edge deployment ready frontend that handles user inputs and visualizes the complex multi–target impacts of the AI recommendations.

Tools and Methods Used

- **Frontend–** Streamlit (rapid UI iteration, native Python bindings).
- **Data Structure–** Pandas & NumPy.
- **Machine Learning–**Scikit-Learn (RandomForestRegressor). Random Forest was chosen over deep learning because it trains quickly on tabular data and provides robust Feature Importance metrics out of the box for XAI.
- **Optimization Search–** SciPy (dual_annealing). Dual Annealing was specifically chosen because Random Forest models create non differentiable "step wise" contour surfaces. Standard gradient-descent optimizers get stuck in local minima on these surfaces; Dual Annealing guarantees finding the global optimum across the entire physical bounds of the machines.

MEASURABLE BUSINESS IMPACT-

We engineered an automated test suite (test_optimization_accuracy.py) to systematically map the performance of the Global Optimizer against mathematically randomized, highly inefficient baseline configurations.

On average, the algorithm systematically prescribed settings that shaved **15.34 kWh of wasted energy per batch**.

- **Factory-Scale ROI-**
  In a high-throughput pharmaceutical or chemical plant executing 100+ batches daily, this AI yields over **1,500 kWh in daily energy savings**. This translates directly into massive margin recovery, reduced carbon footprint, and a software payback period measured in weeks, not years.

- **Empirical Accuracy-**
  Over an 80/20 Train-Test split, our AI Model achieved production-grade accuracy in simulating complex outcomes: Total Energy R^2 = 0.94, Tablet Hardness R^2 = 0.89, Content Uniformity R^2 = 0.91.

Potential Real-World Applications

- **Scalability**

  Continuous Feedback Loops–
  The **SurrogateModel** class is decoupled, it can continuously ingest new batch data from a historian database (like PI Server) and automatically retrain itself on the weekend, continually closing the gap on machine drift.

  Edge & Cloud Deployment–
  The architecture is lightweight enough to be packaged into a Docker container and deployed directly onto an Industrial Edge computer near the PLC, or scaled horizontally on the Cloud to optimize hundreds of parallel lines simultaneously.

Potential Real-World Applications

- **Industry Specific Deployments**
  The engine is designed for any multi-objective process

  - **Pharmaceuticals**
    Ensuring rigorous FDA and ISO quality standards like Content Uniformity and Dissolution Rate are met while optimizing energy.

  - **Chemical Processing**
    Managing complex "trilemmas" of sustainability, quality, and throughput in high volume batch production.

  - **Food & Beverage**
    Using the "What-If" simulator to maintain taste profiles (quality) while reducing the carbon footprint of industrial ovens or cooling systems.

Potential Real-World Applications

- **Flexible Deployment Architecture**

  The engine is built to live where the data is, whether that is on the factory floor or in the cloud

Edge Computing–

The lightweight architecture can be packaged into a Docker container and deployed on an Industrial Edge computer directly next to the PLC (Programmable Logic Controller).

Cloud Scalability–

It can also be scaled horizontally in the cloud to optimize hundreds of parallel manufacturing lines simultaneously across different global sites.

Potential Real-World Applications

- **Relevance & Business Value**

This platform directly addresses the "Trilemma" of modern manufacturing: Sustainability, Quality, and Throughput. By saving just 10–20 kWh per batch, high-throughput manufacturing facilities (e.g., 5,000 batches a year) can save tens of thousands of dollars in electricity costs while drastically lowering their Scope 2 Carbon Footprint—all without compromising rigorous FDA/ISO quality standards.

THANK YOU